JINSEONG PARK

jinseong@kias.re.kr | https://jinseongp.github.io/ | jinseongP | scholar

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Summary

Affiliation **Korea Institute for Advanced Study (KIAS)** *Seoul, Republic of Korea*

AI Fellow (AI Assistant Professor), Center for AI and Natural Sciences

Research Topics Private & Safe Intelligence (PSI, Ψ)

Investigating theoretical foundations and practical algorithms for privacy- and safety-preserving AI systems. My research extends to industrial applications by designing scalable and robust systems through trustworthy AI.

Academic Position

Korea University *Seoul, Republic of Korea*

(Incoming) Assistant Professor, School of Cybersecurity Mar 2027 -

Korea Institute for Advanced Study (KIAS) *Seoul, Republic of Korea*

AI Fellow (AI Assistant Professor), Center for AI and Natural Sciences Mar 2026 - Feb 2027

AI Research Fellow, Center for AI and Natural Sciences Mar 2025 - Feb 2026

- Director: Changbong Hyeon, Distinguished Professor: Daniel Lee, June E Huh

University of British Columbia (UBC) *Vancouver, Canada*

Postdoctoral Fellow, Department of Computer Science (host: Mi Jung Park) Jun 2025 - Jul 2025

Education

Seoul National University (SNU) *Seoul, Republic of Korea*

Ph.D. in Industrial Engineering Mar 2022 - Feb 2025

- Thesis: *Generalization Frameworks for Temporal and Private Deep Learning: From Synthetic Data Generation to Robust Optimization* (Supervisor: Jaewook Lee)
- Cumulative Major GPA: 3.94/4.30

M.S. in Industrial Engineering Mar 2020 - Feb 2022

- Thesis: *Differentially Private Multi-class Classification Using Kernel Supports and Equilibrium Points* (Supervisor: Jaewook Lee)
- Cumulative Major GPA: 4.19/4.30

Pohang University of Science and Technology (POSTECH) *Pohang, Republic of Korea*

Bachelor of Science in Industrial and Management Engineering (IME) Mar 2016 - Feb 2020

- Graduated with Honors: Summa Cum Laude (rank: 1/22)
- Cumulative GPA: 3.91/4.30, Cumulative Major GPA: 4.03/4.30
- Student Exchange Program, Technische Universität Graz (TU Graz) Mar 2019 - Aug 2019

Hana High School (Hana Academy Seoul) Mar 2013 - Feb 2016

Publications

Full list available on 🎓Google Scholar. (†: equal contribution. *: corresponding author.)

J: JOURNALS

[7] Hoki Kim, **Jinseong Park**, Woojin Lee, Jaewook Lee, Yujin Choi, "Exploring the effect of multi-step ascent in sharpness-aware minimization". In: *IEEE Access* (2026).

[6] Yujin Choi, **Jinseong Park**, Youngjoo Park, Jaewook Lee, Junyoung Byun, "Differentially private upsampling for enhanced anomaly detection in imbalanced data". In: *Engineering Applications of Artificial Intelligence* (2026), p. 113124.

[5] **Jinseong Park**, Hyungjin Ko, Jaewook Lee, "Modeling Asset Price Process: An Approach for Imaging Price Chart with Generative Diffusion Models". In: *Computational Economics* (2024), pp. 1–27.

[4] **Jinseong Park**, Hoki Kim, Yujin Choi, Woojin Lee, Jaewook Lee, "Fast sharpness-aware training for periodic time series classification and forecasting". In: *Applied Soft Computing* (2023), p. 110467.

[3] Hoki Kim, **Jinseong Park**, Jaewook Lee, "Generating Transferable Adversarial Examples for Speech Classification". In: *Pattern Recognition* 137 (2023), p. 109286.

[2] **Jinseong Park**, Yujin Choi, Junyoung Byun, Jaewook Lee, Saerom Park, "Efficient differentially private kernel support vector classifier for multi-class classification". In: *Information Sciences* 619 (2023), pp. 889–907.

[1] Yujin Choi, **Jinseong Park**, Jaewook Lee, Hoki Kim, "Exploring Diverse Feature Extractions for Adversarial Audio Detection". In: *IEEE Access* 11 (2023), pp. 2351–2360.

C: CONFERENCES

[12] **Jinseong Park**†, Seungyun Lee†, Woojin Jeong, Yujin Choi, Jaewook Lee, "TimeBridge: Better Diffusion Prior Design with Bridge Models for Time Series Generation". In: *SIGKDD Conference on Knowledge Discovery and Data Mining* (2026).

[11] Hoki Kim, **Jinseong Park**, Yujin Choi, Jaewook Lee, "Stability Analysis of Sharpness-Aware Minimization". In: *International Conference on Machine Learning* (2026).

[10] Miso Kim, Georu Lee, Yunji Kim, Hoki Kim*, **Jinseong Park***, Woojin Lee*, "Co-occurring Associated REtained concepts in Diffusion Unlearning". In: *The Fourteenth International Conference on Learning Representations* (2026).

[9] **Jinseong Park**, Yujin Choi, Jaewook Lee, "Multi-Class Support Vector Machine with Differential Privacy". In: *Advances in Neural Information Processing Systems* (2025).

[8] Yujin Choi, Youngjoo Park, Junyoung Byun, Jaewook Lee, **Jinseong Park***, "Safeguarding Privacy of Retrieval Data against Membership Inference Attacks: Is This Query Too Close to Home?" In: *Conference on Empirical Methods in Natural Language Processing (EMNLP) Findings* (2025).

[7] Dongbin Kim, **Jinseong Park**, Jaewook Lee, Hoki Kim, "Are Self-Attentions Effective for Time Series Forecasting?" In: *Advances in Neural Information Processing Systems* (2024).

[6] **Jinseong Park**†, Yujin Choi†, Jaewook Lee, "In-distribution Public Data Synthesis with Diffusion Models for Differentially Private Image Classification". In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition* (2024).

[5] Yujin Choi†, **Jinseong Park**†, Hoki Kim, Jaewook Lee, Saerom Park, "Fair Sampling in Diffusion Models through Switching Mechanism". In: *Proceedings of the AAAI Conference on Artificial Intelligence* (2024).

[4] Hoki Kim, **Jinseong Park**, Yujin Choi, Jaewook Lee, "Fantastic Robustness Measures: The Secrets of Robust Generalization". In: *Advances in Neural Information Processing Systems* (2023).

[3] **Jinseong Park**, Hoki Kim, Yujin Choi, Jaewook Lee, "Differentially Private Sharpness-Aware Training". In: *International Conference on Machine Learning* (2023).

[2] Sungyoon Lee, **Jinseong Park**, Jaewook Lee, "Implicit Jacobian regularization weighted with impurity of probability output". In: *International Conference on Machine Learning* (2023).

[1] Sungyoon Lee, Woojin Lee, **Jinseong Park**, Jaewook Lee, "Towards Better Understanding of Training Certifiably Robust Models against Adversarial Examples". In: *Advances in Neural Information Processing Systems* 34 (2021).

W: WORKSHOPS

[4] **Jinseong Park**, Mijung Park, "Not Every Time and Frequency Need to Be Forgotten in Diffusion Unlearning". In: *ICML 2026 Workshop on Foundations of Deep Generative Models (FoGen)* (2026).

[3] Changsu Shin, **Jinseong Park***, Sungyoon Lee*, "Memorization Detection in Diffusion Models via Text Embedding Interpolation". In: *ICML 2026 Workshop on Foundations of Deep Generative Models (FoGen)* (2026).

[2] Hyundo Choi, Junhyeong An, **Jinseong Park***, Jaewoong Choi*, "Unlearning for One-Step Generative Models via Unbalanced Optimal Transport". In: *ICML 2026 Workshop on Foundations of Deep Generative Models (FoGen)* (2026).

[1] Seungju Lee, **Jinseong Park**, Jaewook Lee, "Enhancing non-linear asset volatility forecasting models with investor sentiment and explainable AI". In: *Korea Computer Congress (KCC) 2023, XAI workshop, Best paper awards (2/37)* (2023).

P: PREPRINTS

[7] Seungyun Lee, Jaewook Lee, **Jinseong Park***, "Time Series Constrained Sampling via Refining Trajectory-Level Control". In: (2026). Under review, NeurIPS 2026.

[6] Georu Lee, Seungwon Jeong, Hoki Kim*, **Jinseong Park***, Woojin Lee*, "Machine Unlearning for Masked Diffusion Language Models". In: (2026). Under review, NeurIPS 2026.

[5] **Jinseong Park**, Yujin Choi, Youngjoo Park, Jaewook Lee, "Privacy-Preserving Asymmetric Banach Kernel Encoding for Support Vector Classification". In: (2025). Under review, Pattern Recognition.

[4] **Jinseong Park**, Jaewook Lee, Hyungjin Ko, "Advanced Market-Adaptive Portfolio Insurance: Dynamic Regime Detection Approach". In: *Available at SSRN 4779183* (2024). Resubmission, Economic Modelling.

[3] Yujin Choi, **Jinseong Park**, Youngjoo Park, Jaewook Lee, "Multiple Hyperplanes with Dual Encoding Support Vector Machine for Industrial Anomaly Detection". In: (2025). Under review, IEEE Transactions on Industrial Informatics.

[2] Yujin Choi, **Jinseong Park**, Junyoung Byun, Jaewook Lee, "Leveraging Programmatically Generated Synthetic Data for Differentially Private Diffusion Training". In: (2024). Under review, Pattern Recognition.

[1] Hoki Kim, **Jinseong Park**, Yujin Choi, Seungyun Lee, Jaewook Lee, "BayesNAM: Leveraging Inconsistency for Reliable Explanations". In: *arXiv preprint arXiv:2411.06367* (2024). Under review, Pattern Recognition.

Patents

[4] **10-2025-0143602, Patent Application, Korea** Oct 2025

- 시계열 예측 인공지능 모델 학습 방법 및 장치, 이를 이용한 시계열 예측 방법 및 장치 (METHOD AND APPARATUS FOR TRAINING ARTIFICIAL INTELLIGENCE MODEL FOR TIME SERIES FORECASTING, METHOD AND APPARATUS FOR TIME SERIES FORECASTING USING THE SAME)

[3] **10-2023-0087024, Patent Application, Korea** Jun 2023

- 적대적 음성 예제를 감지 방법 및 시스템 (METHOD AND SYSTEM FOR DETECTING ADVERSARIAL SPEECH EXAMPLE)

[2] **10-2023-0087006, Patent Application, Korea** Jul 2023

- 적대적 음성 예제 생성 방법 및 시스템 (METHOD AND SYSTEM FOR GENERATING ADVERSARIAL SPEECH EXAMPLE)

[1] **10-2023-0073178, Patent Application, Korea** Jul 2023

- 커널 서포트 벡터에 기반한 차분 프라이버시 다중 분류 학습을 위한 장치 및 방법 (Apparatus and Method for Learning Differential Privacy Multi-Classification Based on Kernel Support Vector)

Invited Talks

Machine Unlearning in Generative AI

- International Conference for the 80th Anniversary of the Korean Mathematical Society (KMS), Seoul, Korea [link] Jun 23, 2026

Data Privacy in (Generative) AI

- Artificial Intelligence Graduate School, UNIST, Ulsan, Korea Mar 18, 2026
- School of Cybersecurity (PQSnC Lab Seminar), Korea University, Seoul, Korea Feb 23, 2026
- KIAS Center for AI and Natural Sciences Winter Workshop, Goseong, Korea [link] Jan 05, 2026
- National Institute for Mathematical Sciences (NIMS), Seongnam, Korea [link] Nov 26, 2025
- Department of Computer Science, Hanyang University, Seoul, Korea [link] Nov 12, 2025

Trustworthy Generative AI

- KIAS Center for AI and Natural Sciences Workshop, Jeongseon, Korea May 30, 2025
- Korean Society for Industrial and Applied Mathematics (KSIAM), Seoul, Korea May 17, 2025

Privacy-Preserving Deep Learning

- Department of Computer Science (Park Lab ML Seminar), UBC, Vancouver, Canada Jul 23, 2025
- Department of Industrial Engineering, UNIST, Ulsan, Korea Jun 11, 2025
- KIAS School of Computational Sciences Workshop, Gangneung, Korea May 22, 2025
- KIAS Center for AI and Natural Sciences Seminar, Seoul, Korea Mar 23, 2025
- PET(Privacy-Enhancing Technologies) MEETUP, Desilo, Seoul, Korea [link] Jul 2024

Teaching Experiences

Lecturer, Soongsil University Fall 2024

- Analysis of Financial Time Series (금융시계열분석) (2024-2, 2150052601, 3 credits), Seoul, Korea
- Teaching evaluation score: **96.4/100.0** (average: 88.3/100.0)

Teaching Assistant (TA), Industrial Engineering (Prof. Jaewook Lee), SNU 2020 - 2023

- Undergraduate courses: Statistics for Industrial Engineering (Spring 2020, Spring 2022, and Spring 2023), Mathematical Methods for Industrial and Management Engineering (Fall 2020)
- Graduate courses: Data Mining Technology (Fall 2022), Advanced Topics in Statistical Learning (Spring 2023)

TA & Instructor, Corporate Education Programs, SNU 2020 - 2023

- Instructor for big data analysis with Python and deep learning with PyTorch in Woori Bank (2022)
- Instructor for AI practice course in Korea Institute of Startup & Entrepreneurship Development (KISED) (2022)
- TA for Machine Learning and Deep Learning for Prof. Jaewook Lee in SNU Big-Data FinTech course (2021 and 2023)
- TA for Python programming for Prof. Namhyung Kim in SNU Big-Data FinTech course (2022)
- TA for Optimization and Linear Algebra for Prof. Jaewook Lee in Samsung Data Scientist for Device Solution (2020-2022) and Industrial Bank of Korea (2020 and 2021)

Academic Experiences

Academic Projects

- (**PI**) Development of Generative Models for Safe and Trustworthy AI (안전하고 신뢰가능한 인공지능을 위한 생성 모델 개발), Korea Institute of Advanced Study (KIAS), Ministry of Science and ICT Mar, 2025 - Present

- (Research assistant) Physical AI의 Physical World 정렬을 위한 생성모델 제어 이론 연구실 Jul, 2026 - Jun, 2027

Workshop Organizer

- KIAS Center for AI and Natural Sciences Winter Workshop, Goseong, Korea [link] Jan 05-08, 2026

Tutorial Lecture

- The 17th KIAS CAC Summer School on Artificial Intelligence & Parallel Computing, Seoul, Korea [link] Jun 22-26, 2026

KIAS CAINS Seminar Hosts

- Seong Joon Oh (KAIST AI), Deploying general AI in the private world [link] May 06, 2026
- Mingyu Kim (Kookmin University), Safety-Guided Flow (SGF): A Unified Framework for Negative Guidance in Safe Generation [link] Apr 17, 2026
- Sangwoong Yoon (UNIST), Generative Modeling and Reinforcement Learning: Energy, Reward, and Value [link] Apr 09, 2026
- Jaewook Lee (Seoul National University), Stability Analysis Toward Safe Physical AI [link] Mar 20, 2026
- Taehyun Hwang (Seoul National University), Contextual Combinatorial Bandits with Nonlinear Function approximation [link] Mar 04, 2026
- Kyungwoo Song (Yonsei University), Statistical Methods for Improving LLM Performance [link] Feb 20, 2026
- Albert No (Yonsei University), What Does It Mean for an LLM to Forget? Recent Progress in Machine Unlearning [link] Feb 11, 2026
- Byeonghu Na (KAIST), Post-training Alignment for Probabilistic Generative Models [link] Jan 14, 2026
- Jaemoo Choi (Georgia Tech), Efficient and Scalable Diffusion Sampling from an Optimal Transport Perspective [link] Jan 12, 2026
- Hoki Kim (Chung-Ang University), Stability analysis of sharpness-aware minimization [link] Sep 24, 2025
- Jongheon Jeong (Korea University), Safety Issues in Generative AI at Scale [link] Sep 11, 2025
- Lai, Chieh-Hsin (Sony AI), Taming Diffusion Models: Controllable Generation and Fast Sampling [link] Sep 03, 2025

Academic Reviewers

- Journal: Pattern recognition, IEEE Internet of Things, TKDE
- Conference: CVPR, NeurIPS, ICML, ICLR, AAAI, KDD

Attendee, Algorithmic Fairness in Graduate Summer School, LA, USA Jul 2022

- Lectured by Cynthia Dwork et al., Institute for Pure & Applied Mathematics (IPAM) at UCLA

Lab Research Intern, IDEA Lab, Pohang, Korea Sep 2019 - Dec 2019

- Industrial Data Engineering & Analytics Lab, hosted by Prof. Hyunbo Cho at POSTECH

Industrial Experiences

[Project] COSMAX, Seoul, Korea Dec 2023 - Feb 2024, Jun 2024 - Sep 2024

- Lead Researcher, Prediction and Generative Neural Networks for Color Cosmetics

[Intern] SK Telecom, Seoul, Korea Jun 2018 - Aug 2018

- R&D Digital Transformation Team Intern
- Developed services for R&D Digital Transformation of internal data clusters

[Project] LNG industry, POSCO Energy, Pohang, Korea 2018

- Student Researcher, with Prof. Ribin Seo at POSTECH
- Organized LNG value chains and strategic implications of POSCO Energy

Scholarship

2023	The 15th Lee Joong Han Award (Research), Industrial Engineering, SNU.
2022 - 2023	Teaching Assistant scholarship, SNU, Full tuition for all semesters.
2021 Fall	Scholarship for Academic Excellence, SNU, Merit-based, 30% of tuition for a semester.
2019 Spring	Scholarship for Exchange Students, POSTECH.
2018 Fall, 2019 Fall	Mentor Scholarship for IME Information System Tech, C++ programming, POSTECH.
2018 - 2019	National Scholarship for Science and Engineering, Republic of Korea, Merit-based, full tuition for all semesters.
2016 - 2017	Scholarship for Academic Excellence, POSTECH, Merit-based, full tuition for all semesters.

References

Jaewook Lee Professor of Industrial Engineering, Seoul National University
1 Gwanak-ro, Gwanak-gu, Seoul, KR
jaewook@snu.ac.kr | https://safeai.snu.ac.kr

Saerom Park Assistant professor of Industrial Engineering, UNIST
50, UNIST-gil, Eonyang-eup, Ulju-gun, Ulsan, KR
srompark@unist.ac.kr | https://sites.google.com/view/safe-ai-lab

Sungyoon Lee Assistant professor of Computer Science, Hanyang University
IT/BT Building, Wangsimni-ro 222, Seongdong-gu, Seoul, KR
sungyoonlee@hanyang.ac.kr | https://sites.google.com/view/sungyoon-lee